



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
COMPUTER ENGINEERING
THIRD SESSIONAL
SUBJECT: (23CE422) DISCRETE MATHEMATICS

Examination : B.Tech. Semester – IV
Date : 18th March 2025
Time : 12 noon to 1:15 p.m.

Seat No. :
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
 2. The symbols used carry their usual meanings.
 3. Assume suitable data, if required & mention them clearly.
 4. Draw neat sketches wherever necessary.
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Q.1 Do as directed. [12]

- CO6 U (a) Pick a number from a subset that consists only of numbers which, when turned upside down, also yield a valid two-digit number; What will be the probability of getting such outcome while rolling two dice at once? [2]
- CO6 A (b) How can you predict given file compression using Huffman coding will benefit about a threshold or not? [2]
- CO6 N (c) In mathematics, RANDOM FIBONACCI SERIES is a stochastic analogue of the Fibonacci sequence defined by the recurrence $f_n = f_{n-1} \pm f_{n-2}$ where the signs + or – are chosen at random with equal probability $\frac{1}{2}$, independently for different n. If the signs alternate in minus-plus-plus-minus-plus-plus-... pattern, the resulting sequence is not actually RANDOM FIBONACCI SERIES. Justify. [2]
- CO3 A (d) Determine the planarity of $K_{3,3}$. [2]
- CO3 N (e) There are eight bridges on a river on an island. A monk has built a house as denoted 'H' in the Figure. There is also a temple 'T' as shown on the island. The monk wants to reach T from H by crossing each bridge only once without missing any bridge. Is it possible? Justify your answer. [2]



- CO3 U (f) Justify: A cut-set and any spanning tree must have at least one edge in common. [2]

Q.2 Attempt *Any THREE* from the following questions. [12]

- CO6 A (a) Suppose you have a six-sided die, and you want to find the probability of rolling a total of 8 when rolling the die twice. Find the answer without exhaustively considering outcomes manually. [4]

- CO6 A (b) How will you better determine the probability that it will rain tomorrow (Event A) based on whether it's cloudy today (Event B)? The weather forecast says there is a 30% of chance of rain tomorrow. Historically, it's been cloudy 70% of the time on rainy days. Overall, it's cloudy 40% of the time. [4]
- CO6 A (c) How can you use mutual information to figure network speed with respect to ongoing uploads and downloads activities? [4]
- CO6 A (d) Define a discrete random variable about result of exam having marks out of 36 and then find out probability function (probability mass function) as well as distribution function (cumulative distribution function) for value 12, assuming uniform distribution. [4]

Q.3 Attempt the following questions. [12]

- CO2 A (a) Is $(R, \blacksquare)$ a group? $R = \{r_0, r_{60}, r_{120}, r_{180}, r_{240}, r_{300}\}$ and $\blacksquare$ be the operation defined by $r_{\theta_1} + r_{\theta_2} = r_{\theta_1 + \theta_2}$. Here, r_θ denotes rotation of geometric figures drawn on a plane by θ degrees. [4]
- CO2 U (b) Let (G, O) be a group, (H, O) be a subgroup of G and let $g_1, g_2 \in G$, Then prove either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$. [4]
- CO2 N (c) Find all possible subgroups of the group $(G, *)$, where $G = \{1, -1, i, -i\}$. [4]

OR

Q.3 Attempt the following questions. [12]

- CO2 A (a) Is $(Z_6, +_6)$ a group? Provide detailed calculations. [4]
- CO2 U (b) What can you determine about the uniqueness of the identity and inverse element for a group? Justify your answer with required proof. [4]
- CO2 N (c) Is $(Z_p, +_p)$ a cyclic group for $p=5$? If yes, can we conclude that all the elements of Z_p except identity element is a generator. Provide detailed calculations. [4]

Bloom's Taxonomy levels: R-Remembering, U- Understanding, A-Applying, N-Analyzing, E- Evaluating, C-Creating